

Chapter 2 Assignment Due date: April 6, 2026 (Mon)

黃丞暘 611211106

2.1

Sol.

i.

$$f(x) = F'(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 1, & \text{o.w.} \end{cases}$$

ii.

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{a+b}{2} \end{aligned}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x^2 dx \\ &= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3} \end{aligned}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2} \right)^2 = \frac{(b-a)^2}{12}$$

- iii.
- The pdf is constant on $[a, b]$, so all values in the interval are equally likely.
 - The graph of pdf is a rectangle over the interval $[a, b]$.

iv.

$$P(7 \leq X \leq 9) = \frac{9-7}{10-5} = \frac{2}{5} = 0.4$$

2.6

Sol.

$$X \sim \text{Bin}(n=10, p=0.2)$$

$$f_X(x) = \binom{n}{x} p^x (1-p)^{n-x}$$

i.

$$E(X) = np = 10 \cdot 0.2 = 2$$

$$Var(X) = np(1 - p) = 2 \cdot (1 - 0.2) = 1.6$$

ii.

$$P(3 \leq X \leq 5) = P(X = 3) + P(X = 4) + P(X = 5)$$

$$= \binom{10}{3}(0.2)^3(0.8)^7 + \binom{10}{4}(0.2)^4(0.8)^6 + \binom{10}{5}(0.2)^5(0.8)^5$$

$$\approx 0.3158$$

2.8

Sol.

10, 12, 15, 17, 20, 20, 23, 25, 28, 30, 40, 43, 45, 48, 50, 50, 59, 61, 64, 70.

i. $20 \cdot \frac{1}{4} = 5 \Rightarrow Q_1 = \frac{a_5 + a_6}{2} = \frac{20 + 20}{2} = 20$

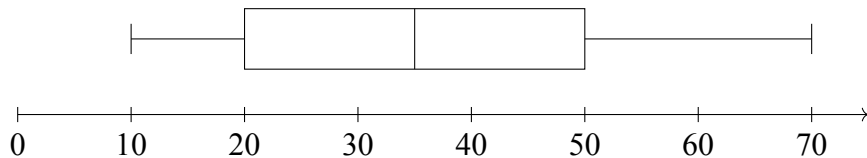
$20 \cdot \frac{1}{2} = 10 \Rightarrow Me = \frac{a_{10} + a_{11}}{2} = \frac{30 + 40}{2} = 35$

$20 \cdot \frac{3}{4} = 15 \Rightarrow Q_3 = \frac{a_{15} + a_{16}}{2} = \frac{50 + 50}{2} = 50$

ii. $IQR = Q_3 - Q_1 = 50 - 20 = 30$

iii. Lower bound = $Q_1 - 1.5IQR = 20 - 45 = -25$

Upper bound = $Q_3 + 1.5IQR = 50 + 45 = 95$



2.9

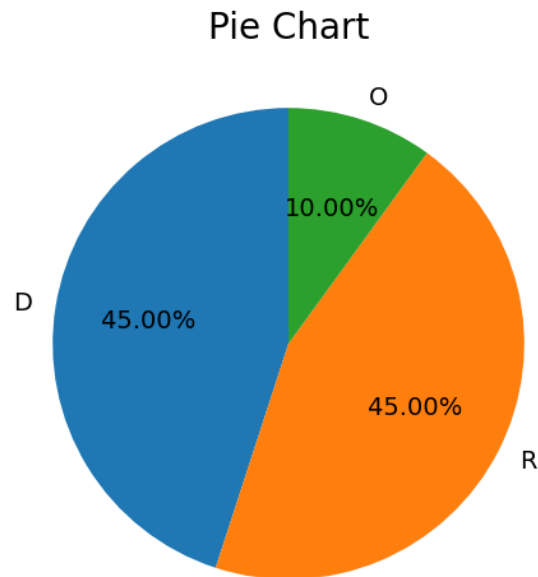
Sol.

D, D, R, O, R, R, R, D, R, D, D, R, D, D, R, R, R, O, D, D

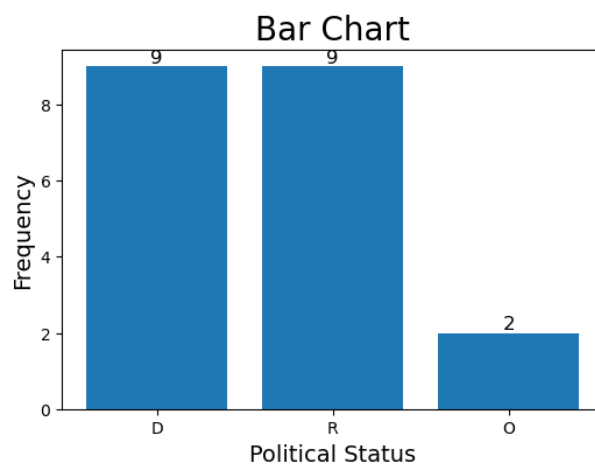
i.

Political Status	Frequency	Relative Frequency
<i>D</i> (Democrats)	9	0.45
<i>R</i> (Republicans)	9	0.45
<i>O</i> (Others)	2	0.10

ii.



iii.



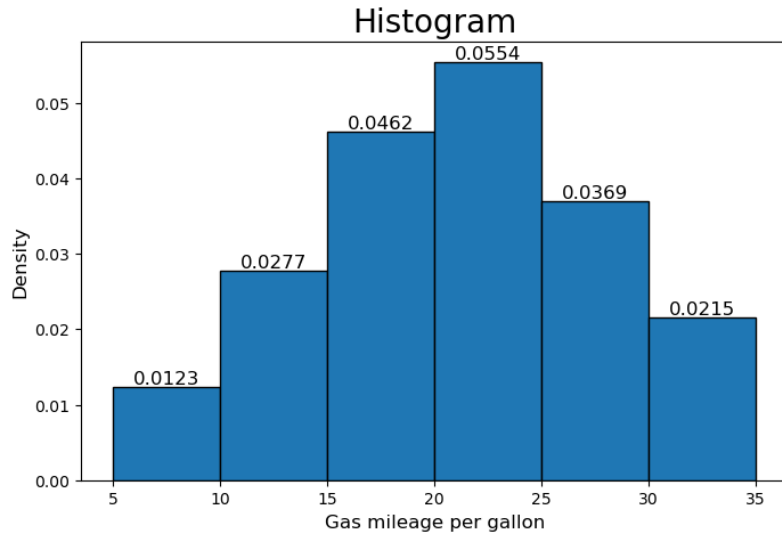
2.11

Sol.

i.

Class Interval	Frequency	Relative Frequency	Density
[5, 10)	20	0.0615	0.0123
[10, 15)	45	0.1385	0.0277
[15, 20)	75	0.2308	0.0462
[20, 25)	90	0.2769	0.0554
[25, 30)	60	0.1846	0.0369
[30, 35)	35	0.1077	0.0215

ii.



iii.

$$P(5 \leq X < 20) = \frac{20 + 45 + 75}{325} \approx 0.4308$$

2.14

Sol.

i.

$$\hat{\pi} = \frac{x}{n} = \frac{272}{400} = 0.68$$

ii. 95% CI for π is

$$\hat{\pi} \pm Z_{0.025} \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{n}} = 0.68 \pm 1.96 \sqrt{\frac{0.68 \cdot 0.32}{400}} = 0.68 \pm 0.0457$$

So the 95% CI is

$$(0.6343, 0.7257)$$

2.19

Sol.

Year	1942	1948	1954	1959	1967	1970	1975	1981	1983
City A	45	0	4	21	9	1	30	17	53
City B	40	10	2	20	7	9	33	17	50
$d_i = A_i - B_i$	5	-10	2	1	2	-8	-3	0	3

i. We test

$$H_0 : \mu_d = 0 \text{ vs } H_1 : \mu_d \neq 0$$

The sample mean of the differences is

$$\bar{d} = \frac{5 - 10 + 2 + 1 + 2 - 8 - 3 + 0 + 3}{9} = -\frac{8}{9} \approx -0.8889$$

and the sample standard deviation is

$$s_d = \sqrt{\frac{\sum(d_i^2) - (\sum d_i)^2/n}{n-1}} = \sqrt{\frac{(5^2 + \dots + 3^2) - [(-8)^2/9]}{8}} \approx 5.1099$$

Thus, the test statistic is

$$t = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}} = \frac{-0.8889}{5.1099/\sqrt{9}} \approx -0.5219$$

The degree of freedom are $9 - 1 = 8$ For a two-sided test with $\alpha = 0.05$, the critical value $t_{0.025, 8} = 2.306$ Since

$$-0.5219 < 2.306$$

So we fail to reject H_0 .

Therefore, at the 0.05 significance level, there is no significance differences between the average snowfalls of the two cities.

2.21

Sol.

5.6, 2.5, 3.7, 7.9, 11.3, 6.4, 3.2, 2.7, 0.5, 5.7, 3.2, 2.4, 7.6, 3.8, 5.7, 5.8, 3.4, 5.1, 2.2, 4.0, 4.2, 4.2, 5.5, 1.5, 1.6, 6.3, 4.3, 5.0, 8.6, 6.3, 4.3, 6.4, 4.6, 2.0, 4.4

i. Let

$$g(\mu) = \mu(\mu + 1) = \theta$$

$$n = 35, \bar{X} = \frac{5.6 + \dots + 4.4}{35} \approx 4.6257, s^2 = \frac{(5.6^2 + \dots + 4.4^2) - [4.6257^2/35]}{35 - 1} \approx 2.2203$$

The point estimate of θ is

$$\hat{\theta} = g(\bar{X}) = \bar{X}(\bar{X} + 1) = 4.6257(4.6257 + 1) \approx 26.0229$$

Since

$$g'(\mu) = 2\mu + 1$$

By delta method, we have

$$\sqrt{n} (g(\bar{X}) - g(\mu)) \rightarrow N(0, [g'(\mu)]^2 \sigma^2)$$

So the approximate standard error is

$$\widehat{SE}(g(\bar{X})) = \sqrt{\frac{(2\bar{X} + 1)^2 s^2}{n}} = \sqrt{\frac{(2 \cdot 4.6257 + 1)^2 2.2203^2}{35}} \approx 3.8473$$

95% CI for θ is

$$\hat{\theta} \pm 1.96 \widehat{SE} = 26.0229 \pm 1.96 \cdot 3.8473 = 26.0229 \pm 7.5407$$

So the 95% CI is

$$(18.4822, 33.5637)$$

```

ii. x = np.array([
    5.6, 2.5, 3.7, 7.9, 11.3, 6.4, 3.2, 2.7, 0.5, 5.7,
    3.2, 2.4, 7.6, 3.8, 5.7, 5.8, 3.4, 5.1, 2.2, 4.0,
    4.2, 4.2, 5.5, 1.5, 1.6, 6.3, 4.3, 5.0, 8.6, 6.3,
    4.3, 6.4, 4.6, 2.0, 4.4])

B = 1000
n = len(x)
np.random.seed(611211106)

def theta_from_mean(m):
    return m * (m + 1)

theta = np.empty(B)

for b in range(B):
    sample = np.random.choice(x, size=n, replace=True)
    mean = np.mean(sample)
    theta[b] = theta_from_mean(mean)

lower = np.percentile(theta, 2.5)
upper = np.percentile(theta, 97.5)

print(f"95% bootstrap CI for theta = ({lower:.4f}, {upper:.4f})")
---
95% bootstrap CI for theta = (19.5137, 34.2572)

```

Using the bootstrap method, the 95% bootstrap CI is

(19.5137, 34.2572)

The delta-method interval is close to the bootstrap interval, and it is slightly shifted to the right.

2.25

Sol.

The kernel density estimator of $f(x)$ is defined by

$$\hat{f}(x) = \frac{1}{n} h_n \sum_{i=1}^n K\left(\frac{x - X_i}{h_n}\right)$$

where $h_n > 0$ is a bandwidth and K is a kernel function. Therefore

```

data = np.array([
    5.6, 2.5, 3.7, 7.9, 11.3, 6.4, 3.2, 2.7, 0.5, 5.7,
    3.2, 2.4, 7.6, 3.8, 5.7, 5.8, 3.4, 5.1, 2.2, 4.0,
    4.2, 4.2, 5.5, 1.5, 1.6, 6.3, 4.3, 5.0, 8.6, 6.3,
    4.3, 6.4, 4.6, 2.0, 4.4])
n = len(data)
h = 1.0

```

```

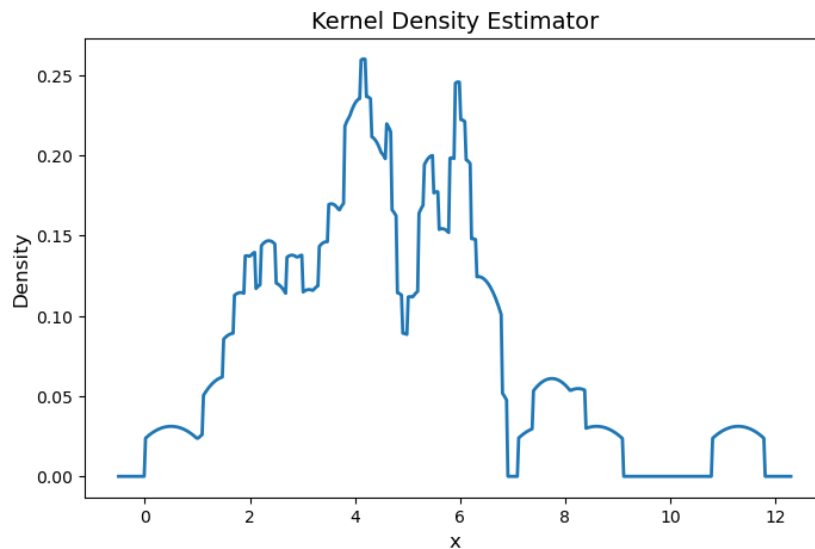
# Epanechnikov kernel
def epanechnikov(u):
    return np.where(np.abs(u)<=0.5, 12/11*(1-u**2), 0.0)

def kde_epanechnikov(x, data, h):
    u = (x[:, None]-data[None, :])/h
    return np.sum(epanechnikov(u), axis=1)/(len(data)*h)

x_grid = np.linspace(min(data)-1, max(data)+1, 500)
y_kde = kde_epanechnikov(x_grid, data, h)

plt.figure(figsize=(8, 5))
plt.plot(x_grid, y_kde, linewidth=2)
plt.xlabel('x', fontsize=12)
plt.ylabel('Density', fontsize=12)
plt.title('Kernel Density Estimator', fontsize=14)
plt.show()

```



2.26

Sol.

i. **bias**

Nadaraya-Waston kernel estimator:

$$\hat{f}_{NW}(x) = \frac{\sum_{i=1}^n Y_i K\left(\frac{x_i - x}{h_n}\right)}{\sum_{i=1}^n K\left(\frac{x_i - x}{h_n}\right)}$$

Assume that nonparametric regression model is

$$Y_i = f(x_i) + \varepsilon_i, \quad i = 1, \dots, n$$

Let

$$w_i(x) = \frac{K\left(\frac{x_i - x}{h_n}\right)}{\sum_{j=1}^n K\left(\frac{x_j - x}{h_n}\right)}$$

then

$$\hat{f}_{NW}(x) = \sum_{i=1}^n w_i(x) Y_i = \sum_{i=1}^n w_i(x) f(x_i) + \sum_{i=1}^n w_i(x) \varepsilon_i$$

Since $E(\varepsilon_i) = 0$, so

$$E[\hat{f}_{NW}(x)] = \sum_{i=1}^n w_i(x) f(x_i)$$

Therefore,

$$Bias(\hat{f}_{NW}(x)) = E[\hat{f}_{NW}(x)] - f(x) = \sum_{i=1}^n w_i(x) f(x_i) - f(x)$$

Use Taylor expansion, we have

$$\begin{aligned} E[\hat{f}_{NW}(x)] &= \sum_{i=1}^n w_i(x) \left[f(x) + f'(x)(x_i - x) + \frac{1}{2} f''(x)(x_i - x)^2 + r((x_i - x)^2) \right] \\ &= f(x) \sum_{i=1}^n w_i(x) + f'(x) \sum_{i=1}^n w_i(x)(x_i - x) + \frac{1}{2} f''(x) \sum_{i=1}^n w_i(x)(x_i - x)^2 \\ &\quad + \sum_{i=1}^n w_i(x) r((x_i - x)^2) \\ &= f(x) + \frac{1}{2} f''(x) h_n^2 \mu_2(K) + r(h_n^2) \end{aligned}$$

where

$$\sum_{i=1}^n w_i = 1, \sum_{i=1}^n w_i(x)(x_i - x) \approx h_n \int u K(u) du = 0, \sum_{i=1}^n w_i(x)(x_i - x)^2 \approx h_n^2 \int u^2 K(u) du = h_n^2 \mu_2(K)$$

Therefore,

$$Bias(\hat{f}_{NW}(x)) = \frac{1}{2} f''(x) h_n^2 \mu_2(K) + r(h_n^2)$$

ii. variance

Since

$$\hat{f}_{NW}(x) = \sum_{i=1}^n w_i(x) f(x_i) + \sum_{i=1}^n w_i(x) \varepsilon_i$$

then

$$Var(\hat{f}_{NW}(x)) = Var\left(\sum_{i=1}^n w_i(x) \varepsilon_i\right) = \sigma^2 \sum_{i=1}^n w_i(x)^2$$

As we know

$$w_i(x) = \frac{K(\frac{x_i - x}{h_n})}{\sum_{j=1}^n K(\frac{x_j - x}{h_n})} \Rightarrow \sum_{i=1}^n w_i(x)^2 = \frac{\sum_{i=1}^n K^2(\frac{x_i - x}{h_n})}{\left[\sum_{j=1}^n K(\frac{x_j - x}{h_n})\right]^2} \approx \frac{nh_n \int K^2(u) du}{[nh_n \int K(u) du]^2} = \frac{nh_n R(K)}{(nh_n)^2} = \frac{R(K)}{nh_n}$$

Therefore,

$$Var(\hat{f}_{NW}(x)) = \frac{\sigma^2 R(K)}{nh_n} + r\left(\frac{1}{nh_n}\right)$$

iii. MSE

Since

$$MSE(\hat{f}_{NW}(x)) = Bias^2(\hat{f}_{NW}(x)) + Var(\hat{f}_{NW}(x))$$

The results from (1) & (2), we have

$$\begin{aligned} MSE(\hat{f}_{NW}(x)) &= \left[\frac{1}{2} f''(x) h_n^2 \mu_2(K) + r(h_n^2) \right]^2 + \left[\frac{\sigma^2 R(K)}{n h_n} + r\left(\frac{1}{n h_n}\right) \right] \\ &= \frac{1}{4} [f''(x)]^2 h_n^4 \mu_2(K)^2 + \frac{\sigma^2 R(K)}{n h_n} + r(h_n^4) + r\left(\frac{1}{n h_n}\right) \end{aligned}$$